Fiscal Year:	FY 2018	**Task Last Updated:**	FY 08/28/2017
PI Name:	Govind, Shubha Ph.D.		
Project Title:	Does Spaceflight Alter the Virulence of a Natural Parasite of Drosophila?		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Developmental biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Animal Biology: Invertebrate		
Space Biology Cross-Element Discipline:	(1) Reproductive Biology (2) Developmental Biology (3) Immunology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
PI Email:	sgovind@ccny.cuny.edu	**Fax:**	FY 212-650-8585
PI Organization Type:	UNIVERSITY	**Phone:**	212-650-8476
Organization Name:	City College of New York		
PI Address 1:	Biology Department		
PI Address 2:	160 Convent Ave		
PI Web Page:			
City:	New York	**State:**	NY
Zip Code:	10031-9101	**Congressional District:**	13
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2014 Space Biology Flight NNH14ZTT001N
Start Date:	11/01/2014	**End Date:**	10/31/2019
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA ARC
Contact Monitor:	Sato, Kevin	**Contact Phone:**	650-604-1104
Contact Email:	kevin.y.sato@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 10/31/2019 per F. Hernandez/ARC (Ed., 4/13/18) NOTE: End date changed to 10/31/2018 per F. Hernandez/ARC (Ed., 9/21/17) NOTE: End date changed to 10/31/2017 per NSSC information (Ed., 8/3/16)		
Key Personnel Changes/Previous PI:	None		
COI Name (Institution):	Bhattacharya, Sharmila Ph.D. (NASA Ames Research Center)		
Grant/Contract No.:	NNX15AB42G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The seminal discovery of Toll and Toll-NF-KB signaling as a central mediator of innate immune responses in the model insect Drosophila catalyzed similar discoveries in other organisms including innate immunity in humans. These findings led to an appreciation of the fundamental roles of innate immunity in human immune deficiency on the one hand, and immune health of insects on the other. In Drosophila larvae, primed, but quiescent hematopoietic progenitors are called into action from an organ with functional similarities to mammalian bone marrow. Blood cells signal the fat body (equivalent to the mammalian liver) to activate the expression of antimicrobial peptide genes (humoral immunity). Antimicrobial peptides efficiently combat deadly bacteria and fungi. The Leptopilina wasps are natural metazoan enemies of Drosophila and their attack kills fruit flies. Wasp attack (e.g., L. boulardi) activates both the humoral and cellular immune arms of larval hosts, each being dependent on Toll-NFKB signaling. Previous NASA-funded studies of Drosophila have shown that stresses in the space environment compromise hematopoietic development and modulate larval and adult innate immunity. We hypothesize that stimuli from the space environment will also alter the virulence of the parasitic wasps, and therefore their success on the fly hosts cultured in space. In Drosophila-wasp co-culture experiments using the NASA Fruit Fly Lab hardware over 30-days, we propose to (1) examine whether the fly or the wasp partner wins the arms race in space; (2) examine the effects of spaceflight on the anti-parasite response in fruit fly hosts; and (3) examine if virulence of "space-cultured" wasps is different from counterparts grown on Earth. While both L. boulardi and L. heterotoma parasites always succeed on the wild type, only the latter wasp species produces unique cell-killing activities residing in its virus-like particles. It also suppresses humoral immunity completely. We will examine if changes in the expression of VLP genes underlies changes in parasite virulence affected by spaceflight. Host defense and virulence will be inferred from a suite of other well-established assays of fly/host survival and measures of immune response.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	We have analyzed the proteome of so-called virus-like particles (VLPs) and found no evidence for a viral nature of VLPs. Instead, the VLP proteome profile appears to possess signature of eukaryotic microvesicles. In addition, we have identified protein domains with prokaryotic character. One particular protein of interest contains structural similarity to bacterial T3 secretion system proteins SipD (in Salmonella) and IpaD (in Shigella) To our knowledge, this is the first kind of an virulence-linked organelle-like entity produced in eukaryotes with elements of secretion systems from both prokaryotes and eukaryotes. Because of the absence of viral character, we have named these structures Multi-Strategy Extracellular Vesicles (or MSEVs). Analysis of several key virulence proteins from MSEVs is underway and their expression and function will be tested in space-flight samples. These studies open new areas of research and strengthen the Drosophila host-parasite system as a stand-in for human and animal studies in space.
Task Progress:	We have analyzed the proteome of so-called virus-like particles (VLPs) and found no evidence for a viral nature of VLPs. Instead, the VLP proteome profile appears to possess signature of eukaryotic microvesicles. In addition, we have identified protein domains with prokaryotic character. One particular protein of interest contains structural similarity to bacterial T3 secretion system proteins SipD (in Salmonella) and IpaD (in Shigella) To our knowledge, this is the first kind of an virulence-linked organelle-like entity produced in eukaryotes with elements of secretion systems from both prokaryotes and eukaryotes. Because of the absence of viral character, we have named these structures Multi-Strategy Extracellular Vesicles (or MSEVs). Analysis of several key virulence proteins from MSEVs is underway and their expression and function will be tested in spaceflight samples. These studies open new areas of research and strengthen the Drosophila host-parasite system as a stand-in for human and animal studies in space.
Bibliography Type:	Description: (Last Updated: 01/09/2024)
Abstracts for Journals and Proceedings	Heavner ME, Ramroop J, Ramrattan G, Dolios G, Crissman J, Scarpati M, Kwiat J, Miller EA, Qiu W, Wang R, Singh S, Govind S. "Functional multi'omics' uncover a novel organelle with both prokaryotic and eukaryotic proteins that weaponizes parasites of Drosophila." 2016 Annual Meeting of the American Society for Cell Biology, San Francisco, CA, December 3-7, 2016. Molecular Biology of the Cell. 2016 Dec 15;27(25):3947. http://dx.doi.org/10.1091/mbc.E16-10-0736 ; Abstract P2422. , Dec-2016
Abstracts for Journals and Proceedings	Jerliu A, Heavne ME, Govind S, Singh S. "Computational modeling and characterization of Diedel proteins that regulate highly-conserved immune signaling." AACR Annual Meeting 2017, Washington, DC, April 1-5, 2017. Cancer Research. 2017 Jul;77(13 Suppl):4537. https://doi.org/10.1158/1538-7445.AM2017-4537 (Proceedings of AACR Annual Meeting 2017.) , Jul-2017
Abstracts for Journals and Proceedings	Heavner ME, Ramroop J, Wey B, Crissman J, Miller EA, Singh S, Govind S. "A new family of GTPases from virulence-linked extracellular microvesicles of a Drosophila generalist parasite." 58th Annual Drosophila Research Conference, San Diego, CA, March 29-April 2, 2017. 58th Annual Drosophila Research Conference, San Diego, CA, March 29-April 2, 2017. Abstract #131. http://www.genetics-gsa.org/drosophila/2017/pdf/Full%20Abstracts%20(including%20late).pdf ; accessed 8/30/17. , Apr-2017
Abstracts for Journals and Proceedings	Govind S, Heavner ME, Ramroop J, Wey B, Singh S. "Molecular analysis of immune-suppressive virus-like particles from cynipid wasp Leptopilina heterotoma, a generalist parasite of Drosophila spp." 2016 XXV International Congress of Entomology, Orlando, FL, September 25-30, 2016. 2016 XXV International Congress of Entomology, Orlando, FL, September 25-30, 2016. Presentation #1777. http://www.entsoc.org/sites/default/files/ICE-2016-Program-Book.pdf , Sep-2016
Articles in Peer-reviewed Journals	Heavner, ME, Ramroop J, Gueguen G, Ramrattan G, Dolios G, Scarpati M, Kwait J, Bhattacharya S, Wang, R, Singh S, Govind S. "Novel organelles with elements of bacterial and eukaryotic secretion systems weaponize parasites of Drosophila." Curr Biol. 2017 Sep 25;27(18):2869-77.e6. Epub 2017 Sep 7. https://doi.org/10.1016/j.cub.2017.08.019 ; PubMed PMID: 28889977 ; PubMed Central PMCID: PMC5659752 , Sep-2017